ML Project Roadmap

Scratch se Machine Learning project kaise banaye - 14 steps

Step 1: Problem choose karo

Pehle decide karo ki kya solve karna hai. Beginner ke liye achhe ideas:

- House price prediction (regression)
- Spam email / SMS detection (classification)
- Movie recommendation system
- Student marks / loan approval prediction

Tip: Aisa topic lo jisme tumhe interest ho aur data easily mil jaye.

Step 2: Basics ready karo

- **Python:** variables, loops, functions, OOP basics
- **Libraries:** NumPy, Pandas, Matplotlib/Seaborn, Scikit-learn
- **Math (basic):** statistics, probability, linear algebra ka intro

Step 3: Environment setup

- Python install karo, phir Jupyter Notebook ya Google Colab use karo (Colab free hai, setup ki tension nahi)
- Project ke liye virtual environment banao:

```
python -m venv venv
pip install numpy pandas matplotlib seaborn scikit-learn jupyter
```

Step 4: Data collect karo

Sources:

- Kaggle (sabse best for beginners)
- UCI Machine Learning Repository
- Government open data (data.gov.in)
- Web scraping ya APIs

Step 5: Data ko samjho (EDA)

- df.head(), df.info(), df.describe() chalao
- Missing values check karo
- Graphs banao: histogram, correlation heatmap, boxplot
- Samjho ki kaunsa feature target par asar daal raha hai

Step 6: Data cleaning aur preprocessing

- Missing values handle karo (fill ya drop)
- Duplicates hatao
- Outliers dekho
- Categorical data ko encode karo (One-Hot, Label Encoding)
- Numerical features ko scale karo (StandardScaler / MinMaxScaler)

Step 7: Feature engineering

- Naye useful features banao (jaise date se month/day nikalna)
- Bekaar features hata do

Step 8: Data split karo

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

Step 9: Model choose karke train karo

Pehle simple model se start karo:

- **Regression:** Linear Regression, Random Forest Regressor
- **Classification:** Logistic Regression, Decision Tree, Random Forest
- Phir advanced try karo: XGBoost, SVM

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

Step 10: Model evaluate karo

- **Classification:** Accuracy, Precision, Recall, F1-score, Confusion Matrix
- **Regression:** MAE, MSE, RMSE, R^2 score
- Overfitting check karo (train vs test score compare karo)

Step 11: Improve karo

- Hyperparameter tuning (GridSearchCV / RandomizedSearchCV)
- Cross-validation
- Alag models compare karo
- Zyada data ya better features try karo

Step 12: Model save karo

```
import joblib
joblib.dump(model, "model.pkl")
```

Step 13: Deploy karo (optional but impressive)

- **Streamlit** ya **Flask** se simple web app banao
- Deploy karo Streamlit Cloud, Hugging Face Spaces ya Render par

Step 14: Documentation aur GitHub

- Code GitHub par upload karo
- Achha README likho: problem, dataset, approach, results, screenshots
- Ye resume/interview ke liye bahut kaam aata hai

Quick summary flow

Problem > Data > EDA > Cleaning > Features > Split > Train > Evaluate > Tune > Save > Deploy > Document